

BT LookAt: use stock

UBTTask_RotateToFaceBBEntry

IN SHORT

UE 5.8 ships `UBTTask_RotateToFaceBBEntry`. It covers LookAt + TurnInPlace use cases. Use it.

Pattern

1. In your BT graph: right-click → Add Task → "Rotate to Face BB Entry".
2. Configure:
 - `BlackboardKey`: a Vector key (target location) or Object key (target actor).
 - `Precision`: degrees of yaw tolerance to count as facing.
3. Task returns `Succeeded` when the agent's rotation is within `Precision` of the target direction.

The stock task uses the AIController's `SetFocus` / `ClearFocus` mechanism so the pawn's rotation continues smoothly even after the task ends (if you want that, use `UBTService_DefaultFocus` to keep focus while a parent node is active).

Why we didn't ship `LCM_BTTask_LookAt` or

`LCM_BTTask_TurnInPlace`

Both planned variants do exactly what `UBTTask_RotateToFaceBBEntry` already does:

- `LookAt`: rotate-toward-target with interp speed → stock task has built-in smooth interp via the AIController focus system
- `TurnInPlace`: constant-rate rotation toward target → stock task supports configurable interp rate

The LCM Nav3D StateTree variants (`FLcmSTTask_LookAt`, `FLcmSTTask_TurnInPlace`) are shipped because UE 5.8 StateTree still has no stock rotation task, but BT has one since 4.x.

When to write a custom LCM Nav3D rotation task

The only case where stock doesn't work: you need **physics-integrated banking** matching the flying-pawn locomotion style. That's already integrated into `LCM_BTTask_FlyTo`'s rotation phase. If you want LookAt-without-translation but WITH banking, fold it into FlyTo with a tiny step size or use the StateTree variant which does have banking-aware rotation.

For >95% of "face the target" use cases, stock `UBTTask_RotateToFaceBBEntry` is the right answer.

Chapter 12, section 3 of the LCM Nav3D User Manual.

The complete manual is `LCM_Nav3D_Manual.pdf`, in this folder alongside every other document in the set. Start from `README.pdf`.

LCM Nav3D: True Volumetric 3D Navigation for Unreal Engine. Version 2.05 · Unreal Engine 5.8.

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